

SQL PROJECT on

HOSPITAL

MANAGEMENT

SYSTEM

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Batch : B3

Project Title : Hospital Management System

Project Goal : To design and implement a Hospital Management

System that efficiently manages patient, doctor, appointment, treatment, and billing data using SQL for organized storage, retrieval, and analysis of hospital records.

HOSPITAL MANAGEMENT SYSTEM

Database Name : hospital_db

Tables Used in database :

1. Patient

-- 1) Patient Table

```
CREATE TABLE patient (  
    patient_id INT PRIMARY KEY,  
    patient_name VARCHAR(100),  
    age INT,  
    gender VARCHAR(10),  
    city VARCHAR(50),  
    disease VARCHAR(100)  
);
```

```
SELECT * FROM patient;
```

2. Doctor

-- 2) Doctor Table

```
CREATE TABLE doctor (  
    doctor_id INT PRIMARY KEY,  
    doctor_name VARCHAR(100),  
    specialization VARCHAR(50),  
    experience INT  
);
```

```
SELECT * FROM doctor;
```

3. Department

-- 3) Department Table

```
CREATE TABLE department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50),  
    doctor_id INT,  
    FOREIGN KEY (doctor_id)  
    REFERENCES doctor(doctor_id)  
);
```

```
SELECT * FROM department;
```

4. Appointment

-- 4) Appointment Table

```
CREATE TABLE appointment (  
    appointment_id INT PRIMARY KEY,  
    patient_id INT,  
    doctor_id INT,  
    appointment_date DATE,  
    FOREIGN KEY (patient_id)  
    REFERENCES patient(patient_id),  
    FOREIGN KEY (doctor_id)  
    REFERENCES doctor(doctor_id)  
);
```

```
SELECT * FROM appointment;
```

5. Treatment

-- 5) Treatment Table

```
CREATE TABLE treatment (  
    treatment_id INT PRIMARY KEY,  
    patient_id INT,  
    treatment_type VARCHAR(100),  
    treatment_cost INT,  
    FOREIGN KEY (patient_id)  
    REFERENCES patient(patient_id)  
);
```

```
SELECT * FROM treatment;
```

6. Billing

-- 6) Billing Table

```
CREATE TABLE billing (  
    bill_id INT PRIMARY KEY,  
    patient_id INT,  
    total_amount INT,  
    payment_status VARCHAR(20),  
    FOREIGN KEY (patient_id)  
    REFERENCES patient(patient_id)  
);
```

```
SELECT * FROM billing;
```

ER-Diagram (Entity Relationship - Diagram) for Hospital Management System :

Entities :

1. **Patient** : patient_id , name, age, gender, city, disease
2. **Doctor** : doctor_id, doctor_name, specialization, experience
3. **Department** : dept_id, dept_name, doctor_id
4. **Appointment** : appointment_id, patient_id, doctor_id, appointment_date
5. **Treatment** : treatment_id, patient_id, treatment_type, treatment_cost
6. **Billing** : bill_id, patient_id, total_amount, payment_status

Relationships :

patient (1) → (many) appointment

doctor (1) → (many) appointment

patient (1) → (many) billing

doctor (1) → (many) department

patient (1) → (many) treatment

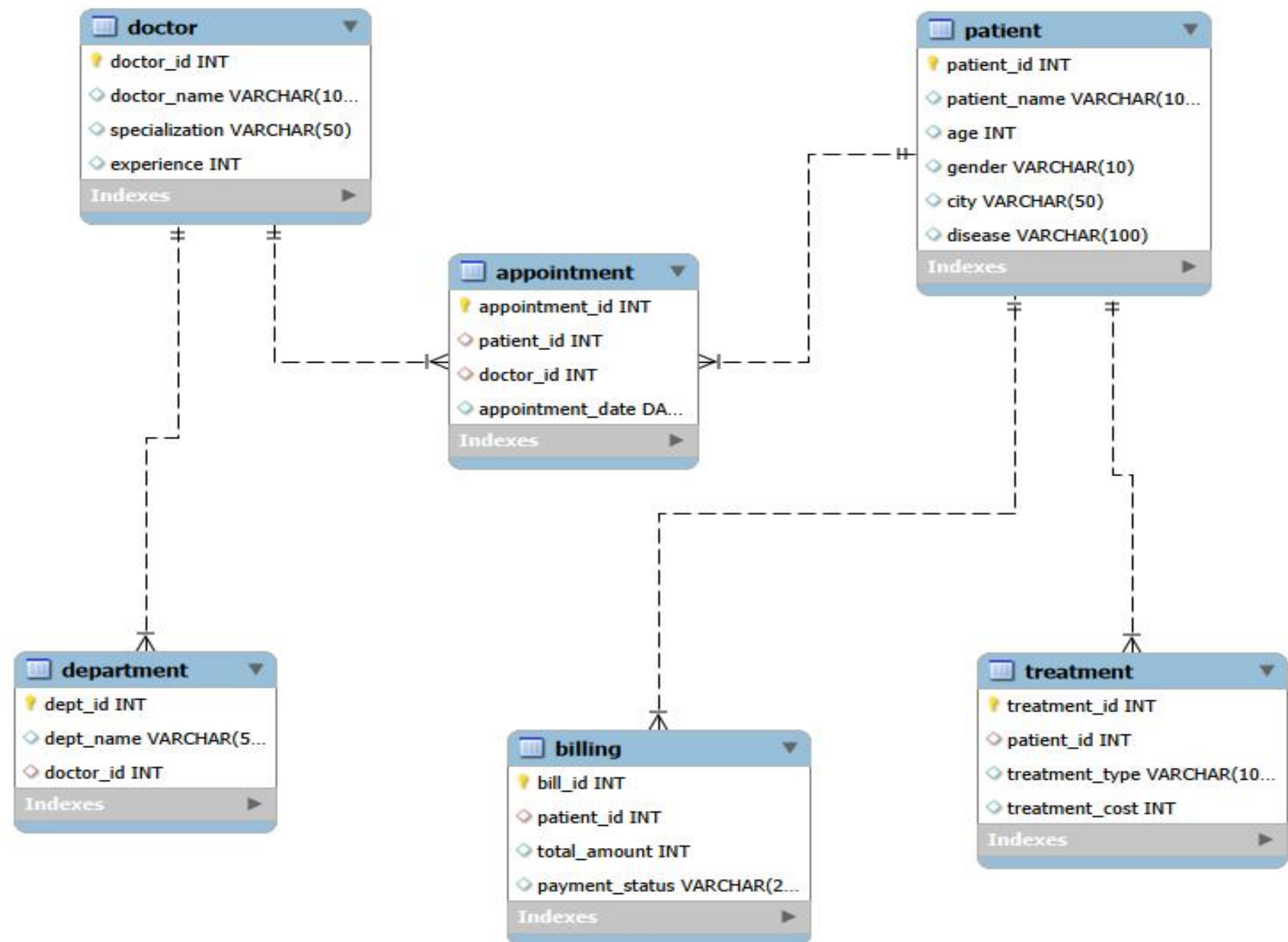


Table Descriptions :

1. Patient :

1	patient_id	patient_name	age	gender	city	disease	
2	1	Patient_1	15	Male	Aurangaba	Hypertension	
3	2	Patient_2	4	Male	Nashik	Hypertension	
4	3	Patient_3	36	Male	Nagpur	Fever	
5	4	Patient_4	32	Male	Kolhapur	Hypertension	
6	5	Patient_5	29	Male	Nagpur	Diabetes	
7	6	Patient_6	18	Female	Nashik	Migraine	

2. Doctor :

1	doctor_id	doctor_name	specialization	experience
2	101	Doctor_10	Pediatricia	21
3	102	Doctor_10	Cardiologi	8
4	103	Doctor_10	Neurologis	26
5	104	Doctor_10	Physician	33
6	105	Doctor_10	Pediatricia	1
7	106	Doctor_10	Dermatolo	35

3. Departments :

1	dept_id	dept_name	doctor_id
2	1	Pediatrics	297
3	2	Neurology	242
4	3	General M	285
5	4	Orthopedi	156
6	5	General M	211
7	6	Cardiology	216

4. Appointments :

1	appointment_id	patient_id	doctor_id	appointment_date
2	1001	84	290	#####
3	1002	44	235	#####
4	1003	159	110	#####
5	1004	118	270	#####
6	1005	177	187	#####
7	1006	93	297	#####

5. Treatments :

1	treatment	patient_id	treatment	treatment_cost	
2	2001	83	Injection	5733	
3	2002	182	Checkup	13784	
4	2003	58	Injection	16966	
5	2004	89	Therapy	10690	
6	2005	133	Checkup	4347	
7	2006	73	Medication	10167	
8	2007	21	Therapy	12543	

6. Billings :

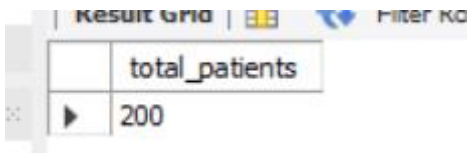
1	bill_id	patient_id	total_amo	payment_status	
2	3001	78	6042	Pending	
3	3002	4	20613	Paid	
4	3003	55	11646	Pending	
5	3004	49	37959	Paid	
6	3005	190	16759	Paid	
7	3006	31	38138	Paid	
8	3007	192	26503	Pending	
9	3008	194	45370	Paid	

Question & Answers :

General Analysis

Que1 : What is the total number of patients in the hospital ?

```
SELECT COUNT(*) AS total_patients  
FROM patient;
```

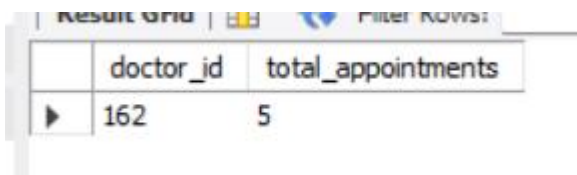


A screenshot of a database result grid. The grid has two columns: 'total_patients' and a value '200'. The grid is titled 'Result Grid' and has a 'Filter Rows' button.

	total_patients
▶	200

Que 2 : Which doctor handled the maximum number of appointments ?

```
SELECT doctor_id, COUNT(*) AS total_appointments  
FROM appointment  
GROUP BY doctor_id  
ORDER BY total_appointments DESC  
LIMIT 1;
```



A screenshot of a database result grid. The grid has two columns: 'doctor_id' and 'total_appointments'. The first row shows '162' and '5'. The grid is titled 'Result Grid' and has a 'Filter Rows' button.

	doctor_id	total_appointments
▶	162	5

Que 3 : Which city has the highest number of patients visiting the hospital ?

```
SELECT city, COUNT(*) AS patient_count  
FROM patient  
GROUP BY city  
ORDER BY patient_count DESC  
LIMIT 1;
```



A screenshot of a database result grid. The grid has two columns: 'city' and 'patient_count'. The first row shows 'Nashik' and '44'. The grid is titled 'Result Grid' and has a 'Filter Rows' button.

	city	patient_count
▶	Nashik	44

Que 4 : Which disease is most commonly treated in the hospital ?

```
SELECT disease, COUNT(*) AS total_cases
FROM patient
GROUP BY disease
ORDER BY total_cases DESC
LIMIT 1;
```

	disease	total_cases
▶	Hypertension	40

Que 5 : Which department has the maximum number of doctors ?

```
SELECT dept_name, COUNT(doctor_id) AS doctor_count
FROM department
GROUP BY dept_name
ORDER BY doctor_count DESC
LIMIT 1;
```

	dept_name	doctor_count
▶	General Medicine	40

Treatment & Cost Analysis

Que 1 : What is the average treatment cost in the hospital ?

```
SELECT AVG(treatment_cost) AS avg_treatment_cost
FROM treatment;
```

	avg_treatment_cost
▶	10707.0300

Que 2 : Which treatment type is most frequently performed ?

```
SELECT treatment_type, COUNT(*) AS frequency
FROM treatment
GROUP BY treatment_type
ORDER BY frequency DESC
LIMIT 1;
```

	treatment_type	frequency
►	Therapy	48

Que 3 : How does treatment cost vary across diseases ?

```
SELECT p.disease, AVG(t.treatment_cost) AS avg_cost
FROM patient p
JOIN treatment t
ON p.patient_id = t.patient_id
GROUP BY p.disease;
```

	disease	avg_cost
►	Hypertension	11170.7442
	Migraine	10413.0606
	Cold	10948.4444
	Fever	11104.9063
	Diabetes	10551.5000
	Infection	9776.3200

Hospital Workflow Analysis

Que 1 : Which patients received treatments costing above average treatment cost ?

```
SELECT patient_id, treatment_cost
FROM treatment
WHERE treatment_cost >
(
SELECT AVG(treatment_cost)
FROM treatment
);
```

	patient_id	treatment_cost
►	182	13784
	58	16966
	21	12543
	142	16169
	33	15408
	161	12742
	78	14126
	158	18497
	137	15928
	24	18113
	152	17719
	149	15139
	40	17807
	44	12354
	160	18986

Que 2 : Which departments handle patients with the highest treatment expenses ?

```
SELECT dept.dept_name, SUM(t.treatment_cost) AS total_cost
FROM department dept
JOIN doctor d
ON dept.doctor_id = d.doctor_id
JOIN appointment a
ON d.doctor_id = a.doctor_id
JOIN treatment t
ON a.patient_id = t.patient_id
GROUP BY dept.dept_name
ORDER BY total_cost DESC;
```

	dept_name	total_cost
▶	Pediatrics	453768
	Orthopedics	414130
	Cardiology	378765
	Dermatology	338346
	Neurology	313574
	General Medicine	256216

Que 3 : Which patients visited the hospital multiple times for appointments ?

```
SELECT patient_id, COUNT(appointment_id) AS visit_count
FROM appointment
GROUP BY patient_id
HAVING COUNT(appointment_id) > 1;
```

	patient_id	visit_count
▶	11	2
	15	2
	20	4
	21	2
	22	2
	24	2
	27	2
	38	2
	39	3
	40	2
	44	2
	45	2
	50	2

Que 4 : Which doctors are handling the highest number of treatments through appointments ?

```
SELECT d.doctor_name as doctor_id , COUNT(t.treatment_id) AS total_treatments
FROM doctor d
JOIN appointment a
ON d.doctor_id = a.doctor_id
JOIN treatment t
ON a.patient_id = t.patient_id
GROUP BY d.doctor_name
ORDER BY total_treatments DESC;
```

	doctor_id	total_treatments
▶	Doctor_162	6
	Doctor_196	6
	Doctor_253	5
	Doctor_213	5
	Doctor_161	4
	Doctor_283	4
	Doctor_292	4
	Doctor_122	4
	Doctor_221	4
	Doctor_207	4
	Doctor_135	4
	Doctor_118	3
	Doctor_235	3

Payment & Revenue Insights

Que 1 : What percentage of total hospital revenue is still pending ?

```
SELECT
SUM(CASE WHEN payment_status='Pending' THEN total_amount
ELSE 0 END)
*100 / SUM(total_amount) AS pending_percentage
FROM billing;
```

	pending_percentage
▶	53.2169

Que 2 : Which patients generated billing amounts above hospital average revenue ?

```
SELECT patient_id, total_amount
FROM billing
WHERE total_amount >
(
SELECT AVG(total_amount)
FROM billing
);
```

	patient_id	total_amount
▶	49	37959
	31	38138
	192	26503
	194	45370
	123	36426
	179	26155
	155	49479
	53	46174
	102	48458
	142	33819
	83	49854
	120	29645
	79	40660
	67	41230

Cross-table Insights

Que 1 : Which diseases require the most expensive treatments overall ?

```
SELECT p.disease, SUM(t.treatment_cost) AS total_cost
FROM patient p
JOIN treatment t
ON p.patient_id = t.patient_id
GROUP BY p.disease
ORDER BY total_cost DESC;
```

	disease	total_cost
▶	Hypertension	480342
	Diabetes	422060
	Fever	355357
	Migraine	343631
	Cold	295608
	Infection	244408

Que 2 : Which city contributes the highest hospital revenue ?

```
SELECT p.city, SUM(b.total_amount) AS revenue
FROM patient p
JOIN billing b
ON p.patient_id = b.patient_id
GROUP BY p.city
ORDER BY revenue DESC
LIMIT 1;
```

	city	revenue
▶	Nashik	1269434

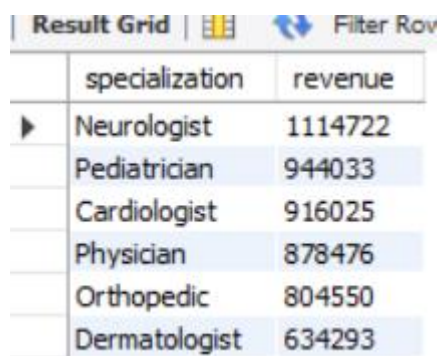
Que 3 : Which doctors treat patients with the highest total treatment costs ?

```
SELECT d.doctor_name as doctor_id, SUM(t.treatment_cost) AS total_cost
FROM doctor d
JOIN appointment a
ON d.doctor_id = a.doctor_id
JOIN treatment t
ON a.patient_id = t.patient_id
GROUP BY d.doctor_name
ORDER BY total_cost DESC;
```

	doctor_id	total_cost
▶	Doctor_253	64031
	Doctor_162	60343
	Doctor_213	53117
	Doctor_196	53088
	Doctor_208	49632
	Doctor_292	48709
	Doctor_122	48193
	Doctor_114	44615
	Doctor_136	43445
	Doctor_161	41552
	Doctor_297	41338

Que 4 : Which specialization contributes the highest hospital revenue ?

```
SELECT d.specialization, SUM(b.total_amount) AS revenue
FROM doctor d
JOIN appointment a
ON d.doctor_id = a.doctor_id
JOIN billing b
ON a.patient_id = b.patient_id
GROUP BY d.specialization
ORDER BY revenue DESC;
```

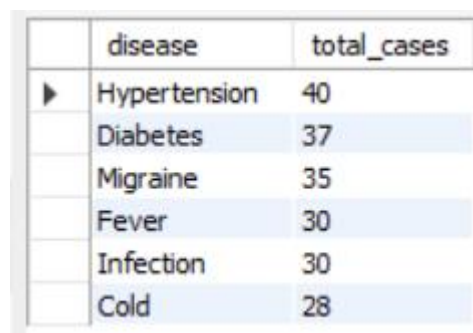


The screenshot shows a 'Result Grid' window with a table containing two columns: 'specialization' and 'revenue'. The data is sorted in descending order of revenue. The rows are: Neurologist (1114722), Pediatrician (944033), Cardiologist (916025), Physician (878476), Orthopedic (804550), and Dermatologist (634293).

	specialization	revenue
▶	Neurologist	1114722
	Pediatrician	944033
	Cardiologist	916025
	Physician	878476
	Orthopedic	804550
	Dermatologist	634293

Que 5 : Which diseases contribute most to hospital workload ?

```
SELECT disease, COUNT(*) AS total_cases
FROM patient
GROUP BY disease
ORDER BY total_cases DESC;
```



The screenshot shows a table with two columns: 'disease' and 'total_cases'. The data is sorted in descending order of total cases. The rows are: Hypertension (40), Diabetes (37), Migraine (35), Fever (30), Infection (30), and Cold (28).

	disease	total_cases
▶	Hypertension	40
	Diabetes	37
	Migraine	35
	Fever	30
	Infection	30
	Cold	28

Conclusion :

The Hospital Management System developed using SQL demonstrates how structured databases can efficiently manage and organize hospital information such as patient details, doctor records, appointments, treatments, and billing data. By applying SQL operations like joins, filtering, grouping, and aggregation, the project enables effective data retrieval and analysis. This system improves data accuracy, reduces manual effort, and supports better decision-making in hospital administration, making it a useful solution for managing healthcare records efficiently.